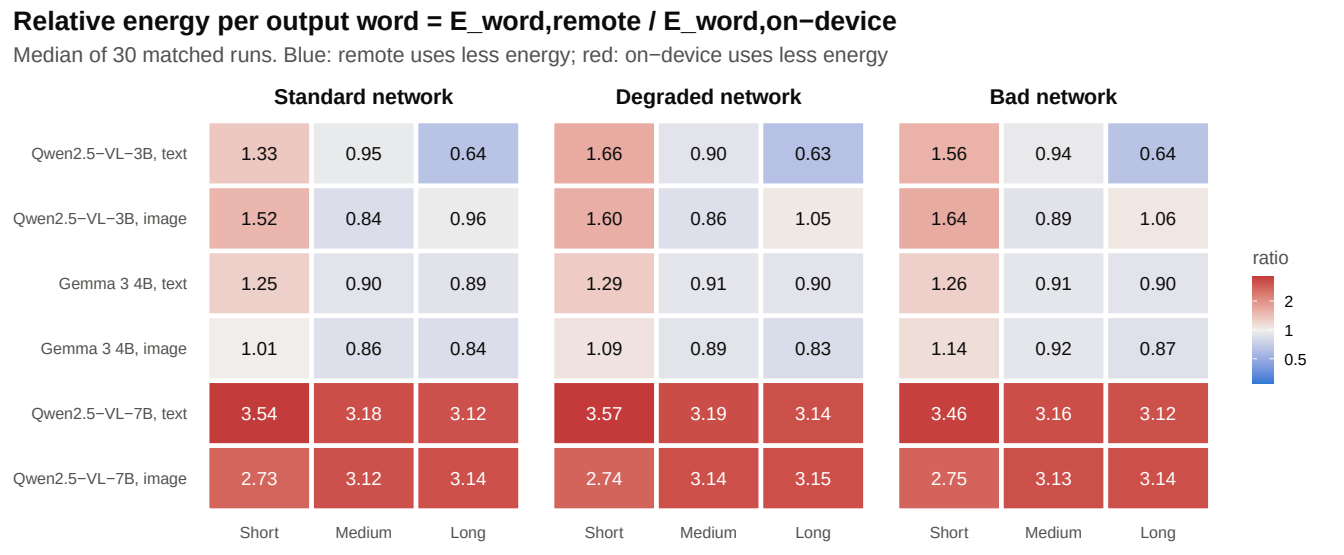
Relative energy per output word = E_word,remote / E_word,on−device
Median of 30 matched runs. Blue: remote uses less energy; red: on−device uses less energy
Standard network
Degraded network
Bad network
Qwen2.5−VL−3B, text
Qwen2.5−VL−3B, image
Gemma 3 4B, text
Gemma 3 4B, image
Qwen2.5−VL−7B, text
Qwen2.5−VL−7B, image
1.33 0.95 0.64
1.66 0.90 0.63
1.56 0.94 0.64
1.52 0.84 0.96
1.60 0.86 1.05
1.64 0.89 1.06
1.25 0.90 0.89
1.29 0.91 0.90
1.26 0.91 0.90
1.01 0.86 0.84
1.09 0.89 0.83
1.14 0.92 0.87
3.54 3.18 3.12
3.57 3.19 3.14
3.46 3.16 3.12
2.73 3.12 3.14
2.74 3.14 3.15
2.75 3.13 3.14
Short
Medium
Long
ratio
2
1
0.5